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ITMO UNIVERSITY

International Research and Educational Center
for Physics of Nanostructures

LAB 4
for the Course
"Adaptive and Robust Control"

on the topic:
OUTPUT FEEDBACK MODEL REFERENCE ADAPTIVE
CONTROL

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INTRODUCTION

0.1 Output Feedback Adaptive Control

Theoretical background. Consider the plant given by the differential equation (or given by the state-space equation from Work 3)

$$y^{(n)} + a_{n-1}y^{(n-1)} + a_{n-2}y^{(n-2)} + \dots + a_0y = b_mu^m + b_{m-1}u^{m-1} + \dots + b_0u$$

where $a_i, i = 0, 1, \dots, n-1, b_j = 0, 1, \dots, m$ are the constant unknown parameters.

We make the following standard assumptions about the plant. Assumption 4.1. The plant is controllable and observable, i.e., the polynomials $a(s)$ and $b(s)$ of the plant transfer function

$$W(s) = \frac{b(s)}{a(s)} = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \dots + a_0}$$

have no common roots. Assumption 4.2. The polynomial $b(s)$ is Hurwitz. Assumption 4.3. The sign of b_m is known. Without loss of generality, we will assume b_m positive.

The objective is to design a control law that will compensate the influence of the plant uncertainties on the closed-loop system stability and ensure the equality

$$\lim_{t \rightarrow \infty} (y_M(t) - y(t)) = 0$$

where $y_M(t)$ is the output of the reference model

$$y_M(t) = \frac{a_{m0}}{s^2 + a_{m1}s + a_{m0}}[g(t)] = W_M(s) [a_{m0}g(t)]$$

in which $g(t)$ is the piece-wise continuous and bounded reference signal,

$$W_M(s) = \frac{a_{m0}}{s^2 + a_{m1}s + a_{m0}}$$

is the reference model transfer function designed using the pole placement (standard polynomials) approach considered in Work 3. We note that the transfer functions $W(s)$ and $W_M(s)$ are not SPR. Hence, an augmented error adaptation algorithm should be applied as one of the solutions to this problem.

0.2 Notations

Throughout this work, the following notations are used:

Table 1 — Table 1: Notations of symbols

Symbol	Explanation
A	Plant model
b_0	Coefficient
t_T	Transient time
e	Control error
x	Output of the plant
x_m	Output of the reference model
$\tilde{\theta}$	Parameter estimation error
g	Reference signal
λ	Parameter for transient performance

0.3 Variant

Table 2 — System parameters and reference signal

No.	Matrix A	Coeff. b_0	Transient time t_T	Reference $g(t)$
15	$\begin{bmatrix} 0 & 1 \\ -15 & -7 \end{bmatrix}$	3	0.7	$\text{sign}(\sin(0.5t)) + 2$

1 SOLUTIONS

The reference model was successfully designed based on the parameters of Variant 15. It exhibits a critically damped response with no overshoot and a transient time of approximately 0.7 s. This model will serve as the desired trajectory for adaptive and non-adaptive control simulations in subsequent tasks.

1.1 Design the tracking adaptive controller

1.1.1 Simulink Model

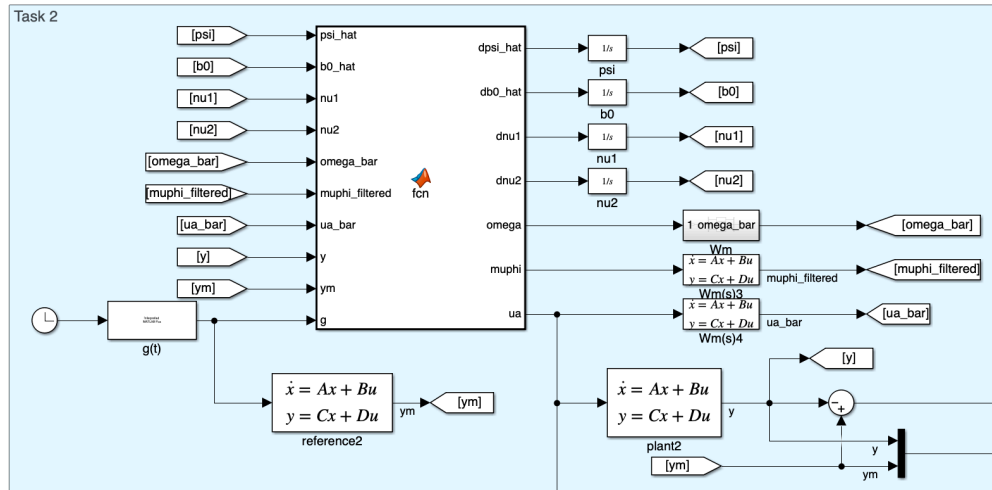


Figure 1 — Simulink implementation of the non-adaptive control system.

Make simulation for three different adaptation gains $\bar{\gamma}_1$ and fixed $\bar{\gamma}_2$.

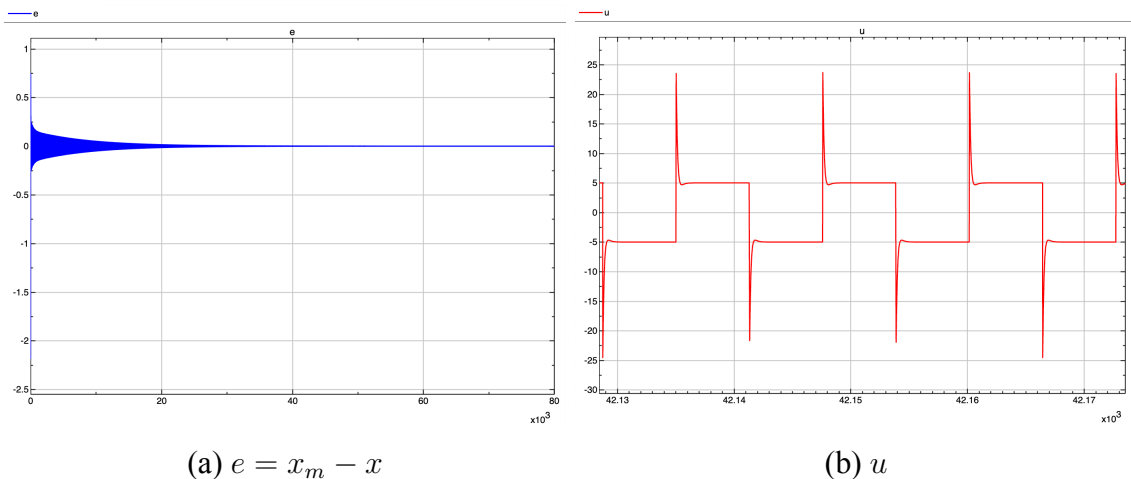
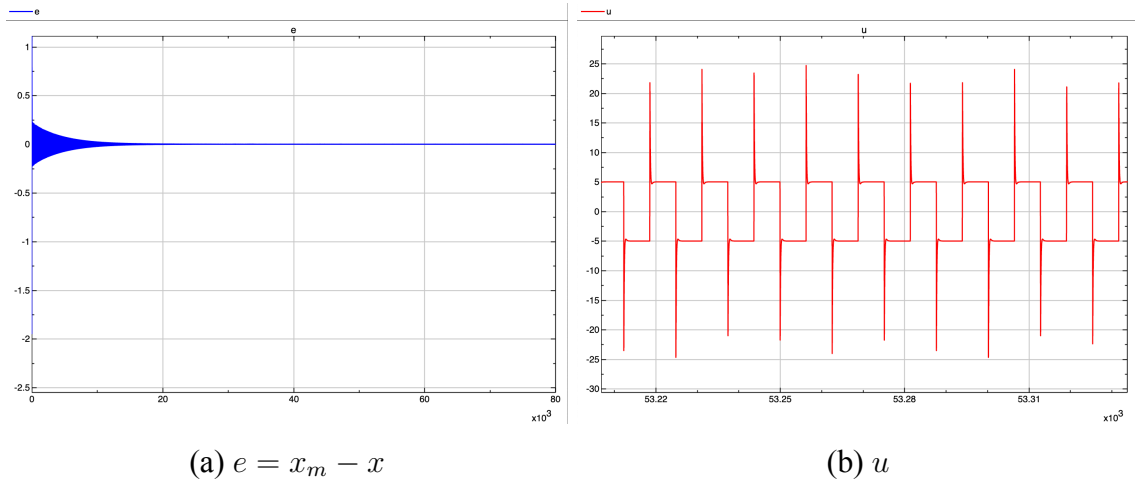
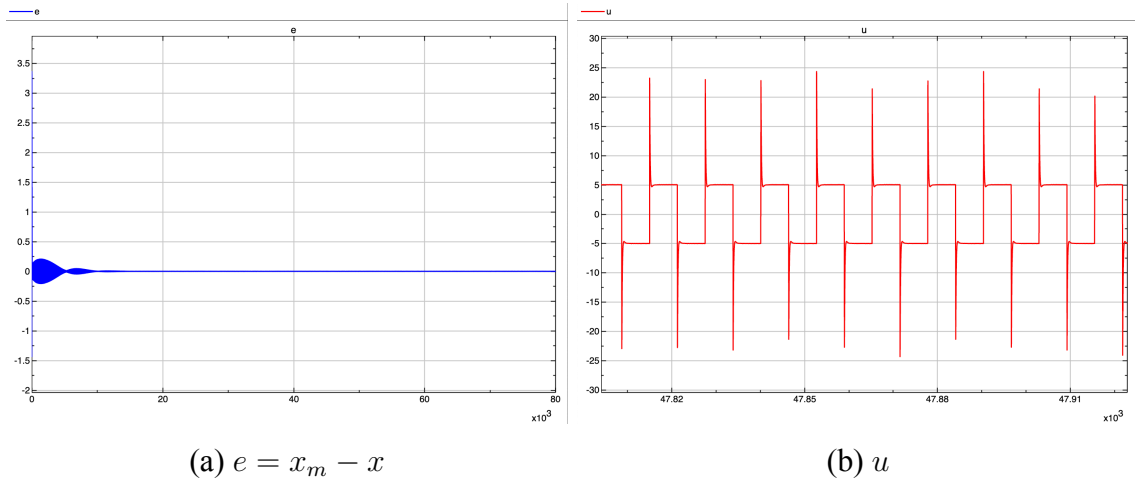


Figure 2 — $\bar{\gamma}_1 = 10$


Figure 3 — $\bar{\gamma}_1 = 20$

Figure 4 — $\bar{\gamma}_1 = 100$

1.1.2 Conclusion

The adaptive controller successfully achieves the primary objective of driving the tracking error $e = x_m - x$ to zero for all adaptation gains. The choice of $\bar{\gamma}_1$ represents a trade-off between convergence speed and control effort.

1.2 Change the plant parameters to unknown

1.2.1 Repeat this triple of experiments with tracking adaptive controller

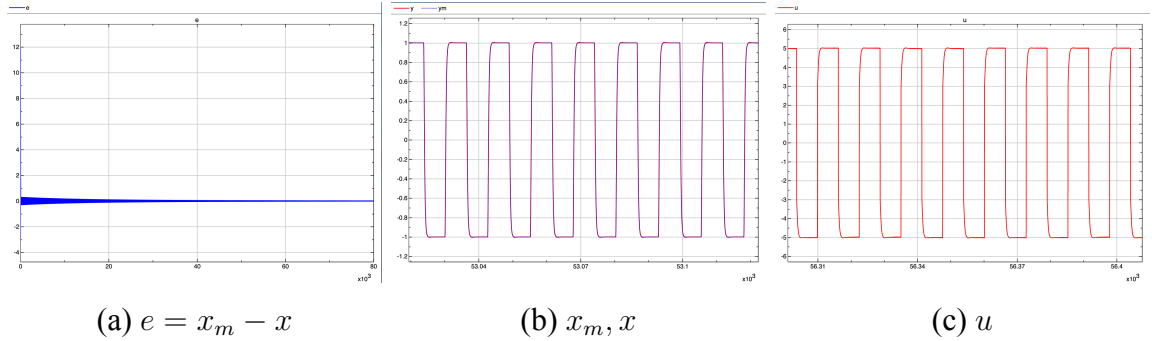


Figure 5 — Apply $A = 2 * A, b = 2 * b$

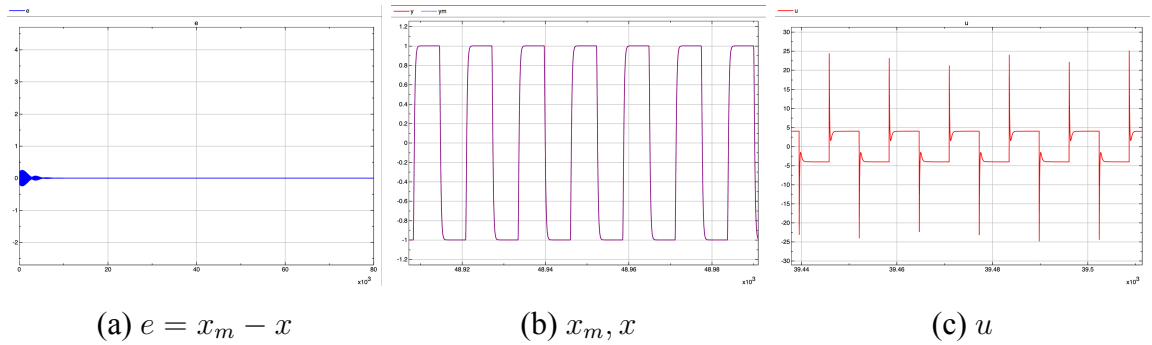


Figure 6 — Apply $A = A + \begin{bmatrix} 0 & 0 \\ 3 & 3 \end{bmatrix}$

1.2.2 Conclusion

In both scenarios, the adaptive control scheme proves its ability to handle unknown plant parameters while maintaining system stability and achieving the control objective $\lim_{t \rightarrow \infty} (x_m(t) - x(t)) = 0$. This validates the theoretical robustness properties of the augmented error approach in output feedback model reference adaptive control.